A Multi-Agent LLM Framework for Explainable Algorithmic Trading on NSE-Listed Equities

MTP Final Evaluation

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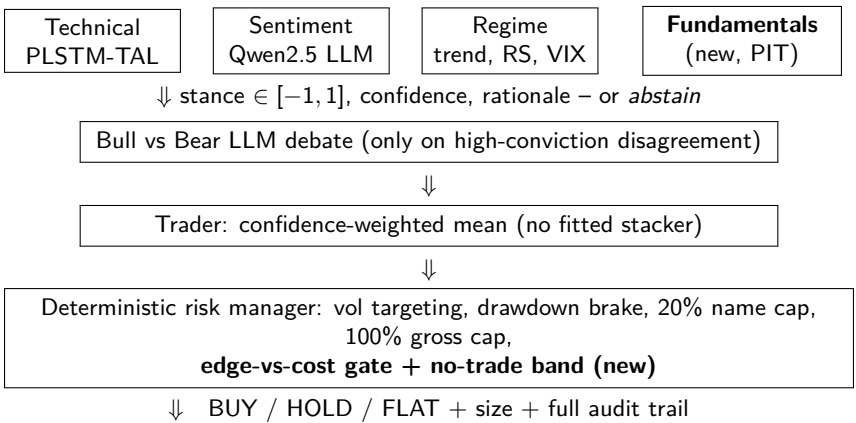
Problem and objectives

- Stock-prediction models give a score with no rationale, tested on US indices, ignoring costs.
- Desks combine specialists – technical, fundamental, sentiment, risk – and debate before trading.
- **Goal**: an explainable multi-agent decision-support system for NSE equities, evaluated out-of-sample under the full Indian cost structure.

Status at final evaluation

- Mid-sem (done): PLSTM-TAL technical agent, LLM sentiment agent, regime agent, debate, risk manager, backtester.
- November plan (done): larger LLM comparison · extended universe + period · earnings-day sentiment · point-in-time fundamentals agent · cost-aware sizing · full ablation re-run · explainability evaluation.

System architecture



Evaluation protocol: built so a negative answer is trustworthy

- Purged, embargoed walk-forward: 3y train / 6m test / 10-day embargo, 16 folds.
- 19,110 out-of-sample decisions, 2018-12-07 – 2026-08-27; decide at close t, trade at open $t+1$.
- Full Indian delivery costs: STT 0.1% each side, stamp duty, NSE/SEBI + GST, 5 bps slippage $\approx$ **32 bps round trip**.
- Block-bootstrap CIs, Holm correction across all configurations, Deflated Sharpe Ratio.
- New components pre-registered: band 0.05, conviction floor 0.10 – never tuned to the result.
- Mid-sem numbers reproduce *exactly* in the final run – new parts are opt-in.

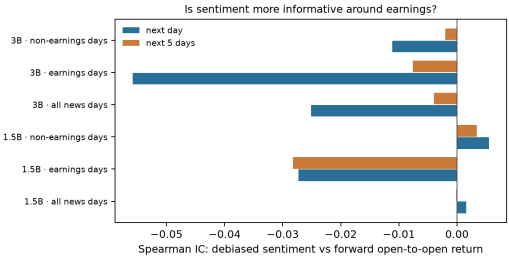
New: point-in-time fundamentals agent

- Annual P&L and balance sheet, 30 names, FY2015–FY2026.
- **No look-ahead**: a report is invisible until FY end + 61 days (SEBI LODR Reg. 33's 60-day deadline).
- Features: profit growth, revenue growth, ROE, ΔROE – z-scored *across peers on each date* using only public reports.
- Handles fiscal-year changes (HCLTECH 9-month 2016, NESTLEIND 15-month 2024) instead of computing fake growth.
- Example rationale: *"FY report Mar 2021 (public since 2021-05-31): return on equity +38.2% is the strongest vs peers (z +2.1) . . . "*
- Limits stated: restated (not as-first-printed) figures; no P/E.

New: sentiment around earnings announcements

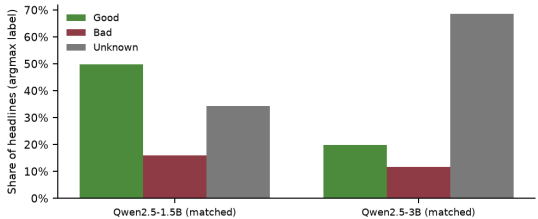
- ∼10% of headlines are results news.
- Earnings days *are* information-rich: |next-day move| 133 vs 126 bps (p=0.029).
- Full corpus: tone does **not** predict direction (1 or 5 days).
- Earnings headlines alone: a Holm-significant **reversal** (IC ≈ -0.085) – good news underperforms next day. Fragile (gone after debiasing), $\approx$ cost-sized: not an edge.

Model	Subset	n	IC	p	Hit rate
Qwen2.5-1.5B	all news days	15190	+0.00	0.84	50.1%
Qwen2.5-1.5B	earnings days	1803	-0.03	0.25	48.7%
Qwen2.5-1.5B	non-earnings days	13387	+0.01	0.53	50.2%
Qwen2.5-3B	all news days	6480	-0.03	0.04	49.2%
Qwen2.5-3B	earnings days	1803	-0.06	0.02	47.9%
Qwen2.5-3B	non-earnings days	4677	-0.01	0.45	49.7%



New: larger LLM (Qwen2.5-3B) vs local 1.5B

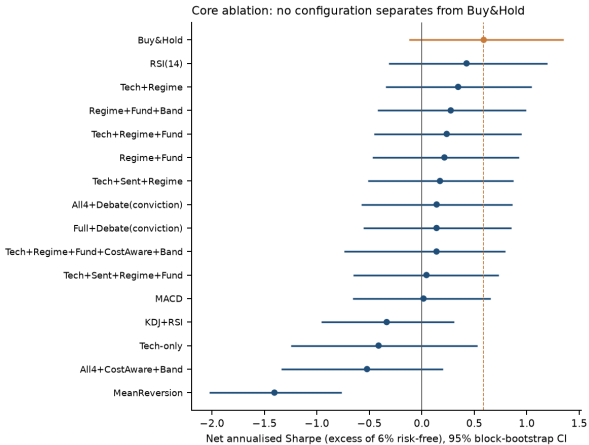
- 1.5B model: Good:Bad = 3.31 – a bullish prior, not bullish news.
- Same 10,240 headlines, same prompt: label agreement 60.3%, $\kappa = 0.39$.
- Qwen2.5-1.5B: IC -0.02 (p=0.13), hit rate 48.4%.
- Qwen2.5-3B: IC -0.03 (p=0.04), hit rate 49.2%.
- 3B is far less bullish and says Unknown twice as often – but neither model gives a usable long-only signal.



Full ablation, core universe (net of costs)

Strategy	Sharpe	Gross	MaxDD	Turnover	Exposure	Holm p
Buy&Hold	0.59	0.59	-38.6%	0×	100%	–
RSI(14)	0.43	0.47	-39.8%	4×	60%	0.82
Tech+Regime	0.35	0.58	-20.7%	20×	86%	0.82
Regime+Fund+Band	0.28	0.47	-22.2%	16×	84%	0.60
Tech+Regime+Fund	0.24	0.50	-23.2%	23×	88%	0.59
Regime+Fund	0.21	0.48	-23.2%	24×	88%	0.59
Tech+Sent+Regime	0.17	0.66	-25.0%	43×	90%	0.41
All4+Debate(conviction)	0.14	0.60	-24.7%	43×	93%	0.09
Full+Debate(conviction)	0.14	0.62	-27.2%	44×	91%	0.21
Tech+Regime+Fund+CostAware+Band	0.14	0.37	-5.3%	5×	7%	0.82
Tech+Sent+Regime+Fund	0.04	0.59	-25.0%	49×	91%	0.14
MACD	0.02	0.46	-29.9%	43×	78%	0.39
KDJ+RSI	-0.34	0.24	-39.3%	51×	34%	0.01
Tech-only	-0.41	-0.27	-25.4%	5×	16%	0.09
All4+CostAware+Band	-0.52	0.20	-5.1%	14×	7%	0.21
MeanReversion	-1.41	0.68	-79.6%	223×	76%	0.00

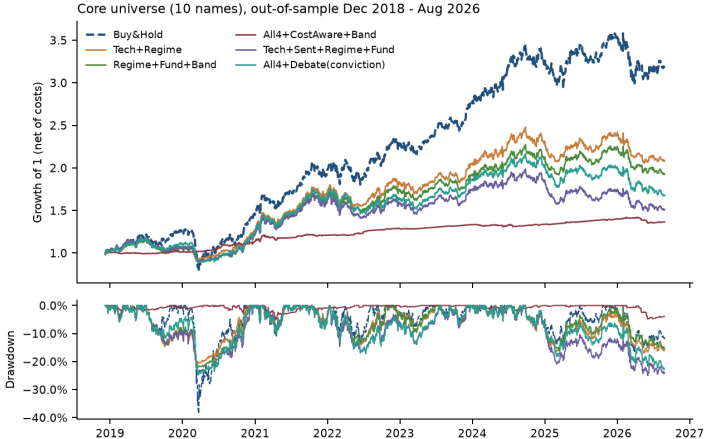
No configuration separates from Buy-and-Hold



What each November component did

- **Fundamentals**: Tech+Regime 0.35 $\rightarrow$ 0.24. Annual rankings carry no daily information.
- **No-trade band**: turnover 24x $\rightarrow$ 16x/yr, Sharpe 0.21 $\rightarrow$ 0.28 – helps by trading less.
- **Edge-vs-cost gate**: almost nothing clears 32 bps – invested 7% of the time. Low drawdown = cash.
- **Conviction-gated debate**: escalates 10.5% of days (mid-sem raw gate: 97.4%; 28.9% with fundamentals); Sharpe 0.14 / 0.14 – LLM cost down $\sim 9\times$, no change in outcome.

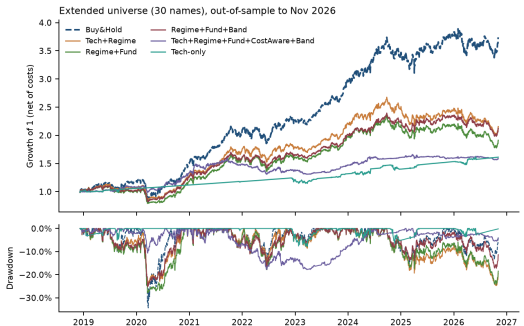
Equity and drawdown: the overlay protects in crashes



Extended: 30 names, data to 2026-11-04

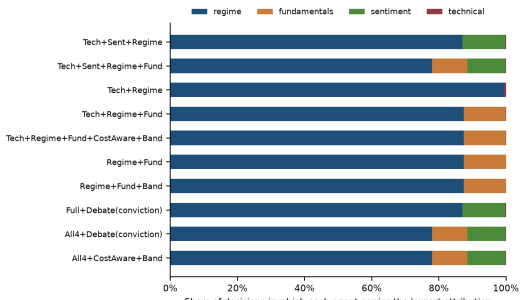
Strategy	Sharpe	MaxDD	Turnover
Buy&Hold	0.74	-34.3%	0×
RSI(14)	0.66	-38.5%	9×
Regime+Fund+Band	0.36	-25.1%	16×
Tech+Regime	0.35	-24.7%	29×
Tech+Regime+Fund	0.25	-27.8%	29×
Regime+Fund	0.25	-27.8%	29×
MACD	0.23	-29.1%	70×
Tech-only	0.09	-7.7%	7×
Tech+Regime+Fund+CostAware+Band	0.06	-17.8%	17×
KDJ+RSI	-0.92	-64.2%	119×
MeanReversion	-1.97	-91.8%	311×

The null replicates on 3× the names and on data the mid-sem never saw.



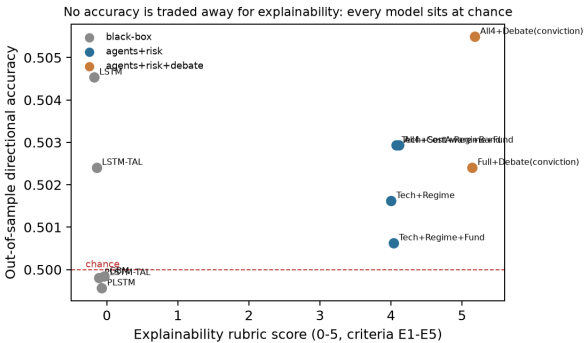
Explainability, measured – not asserted

1. **Provenance**: does the rationale name the model that made the number? Mid-sem logs: **0%** – all 19,110 credited PLSTM-TAL for LSTM-TAL. Caught by this audit; fixed: 100%.
2. **Integrity**: stated stances = logged stances: 100%.
3. **Reconstruction**: score recomputed from opinions: 100%.
4. **Deletion**: removing the top-attributed agent flips 37%–71% of decisions vs 0.0%–3.4% for another agent.



The explanations reveal the system is a trend filter: the regime agent decides most calls.

Accuracy–explainability trade-off: there is nothing to trade



- Black-box LSTMs and GBM: chance.
- Explainable agent systems: chance.
- In a low-signal market the binding constraint is *signal*, not interpretability.
- The explainable design is strictly better: it is the one that shows *why* there is no edge.

Conclusions

- Built and tested end-to-end: 5 agents, debate, risk manager, cost-aware backtester, live paper-trading demo, 350+ automated tests.
- **No configuration beats Buy-and-Hold** after real Indian costs – on 10 names, on 30 names, and on fresh data.
- None of the November additions changes that; the no-trade band helps only by cutting turnover.
- What survives: a regime-aware risk overlay that roughly halves drawdowns, and an explanation layer that is measurably faithful – and diagnosed its own bug.
- Limitations: open 1.5B/3B models only, survivorship, restated fundamentals, power $\approx$ Sharpe 1.0.
- Next: event-time / intraday horizons, vendor PIT fundamentals, frontier LLM debate, market-neutral formulation.

Thank you

Questions?